KAJAL SANWAL

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PROFESSIONAL SUMMARY

Software Engineer | Software Development Engineer | Full-Stack | Backend

Experience building scalable, high-performance applications across full-stack web development and embedded systems. Proficient in React, Next.js, Node.js, PostgreSQL, and REST API development. Strong foundation in C/C++, firmware development, and hardware-software integration. Familiar with microservices architecture, distributed systems basics, and performance optimization. Pursuing BS in Electronic Systems at IIT Madras.

EDUCATION

Degree / Board - Institute Year: GPA / %

BS, Electronic Systems - IIT Madras: 2027 (8.32)

SCHOLASTIC ACHIEVEMENTS

- JEE Main Percentile: 99.1
- LeetCode: Solved 250+ problems (Data Structures & Algorithms); regular participant in weekly and biweekly contests
- Gold Medalist, Maths Olympiad (1st rank at School level)
- Science Olympiad (SOF): Secured olympiad rank 62 for distinctive performance, 2019.
- International Reasoning & Aptitude Olympiad (SilverZone): Achieved Rank 96, 2018.

EXPERIENCE

Co-founder & Full-Stack Developer | SelfShft — June 2026 – Present

- Built a full-stack web platform using Next.js, TypeScript, and modern web technologies to connect students with location-based part-time jobs and short-term gigs.
- Startup selected in the **Stage 1** cohort of IIT Madras Nirmaan.
- Designed and implemented features for student onboarding, employer profiles, job matching, and authentication.
- Collaborated on product strategy and technical architecture as part of the founding team.
- Helped develop a platform addressing financial accessibility for students relocating for higher education.

Embedded Systems / Firmware Intern — Instrumentation Team — BluArmor | Aug 2025 – Oct 2025

Pre-Placement Offer (PPO) extended upon completion

- Designed automated RF/Bluetooth characterization systems using STM32 firmware, relay matrices, and ADC acquisition, reducing manual testing time by 15%.
- Built Python-based calibration and automation workflows, improving production measurement accuracy by 33%.
- Developed Python GUI tools for Bluetooth RSSI testing with automated CSV logging and live visualization, increasing throughput to 500 devices/day.
- Improved RSSI measurement stability by 55% using RF shielding and Exponential Moving Average (EMA) filtering. • Automated headset pairing & validation workflows on Raspberry Pi 5 and nRF52840, enabling parallel multi-device testing.

PROJECTS & SYSTEM DESIGN

1. AI-Powered SaaS Platform | Next.js · Prisma · NeonDB · Clerk · Cloudinary | **Live** | **GitHub**

- Built scalable multi-tenant SaaS system supporting 50+ concurrent tenants with role-based authentication. • Integrated Cloudinary for optimized media uploads and transformations, reducing media load time by ~40%. • Deployed serverless AI workloads with high-availability auto-scaling on Vercel edge infrastructure.

2. Full-Stack Auth & Admin Notes | Node.js · MongoDB · Express · JWT · RBAC | **Live** | **GitHub** •

Built JWT + RBAC authentication system deployed to 200+ active IIT Madras users.

- Designed indexed MongoDB schemas reducing query time by ~30%; implemented secure note approval workflows.

3. VibeCode Editor | React · Zustand · Appwrite · TypeScript | **GitHub**

- Developed a high-performance browser-based IDE with virtualized file explorer and live execution playground, achieving sub-50ms re-render latency.

• Managed complex state with Zustand and utilized Appwrite BaaS for secure database schemas and real-time sync.

4. Fastify Backend & Thumbnailing Service | Fastify · Node.js Streams · JWT · PostgreSQL | **Live** | **GitHub**

• Built scalable backend system using Fastify with modular architecture and Node.js streams for efficient file handling up to 100MB.

• Designed JWT authentication and structured logging; average API response time under 80ms. • Implemented RESTful APIs with optimized latency and structured logging.

TECHNICAL SKILLS

Languages: C/C++, Embedded C, Python, JavaScript, TypeScript, SQL, Linux Shell
Frontend: React, Next.js (App Router), HTML, CSS, Tailwind CSS, shadcn/ui
Backend & DBs: Node.js, Express, Fastify, REST APIs, MongoDB, PostgreSQL (Neon), Prisma ORM
Auth & Security: JWT, RBAC, Clerk, OAuth basics, Secure API Design
Embedded / HW: STM32 HAL, FreeRTOS, PWM, Raspberry Pi 5, nRF52840, PCB design, MATLAB, LTSpice
DevOps / Tools: Git & GitHub, Vercel, CI/CD basics, Docker (familiar), Postman, Cloudinary, VS Code **Core CS:** Data Structures & Algorithms, OOP, System Design Fundamentals, Problem Solving **Concepts:** Scalable Systems, Microservices Architecture, Distributed Systems (Basics), Performance Optimization

POSITIONS OF RESPONSIBILITY

Deputy Head, Anukriti — Paradox, IIT Madras *March 2025*
Teaching Assistant, Embedded C Programming Lab — IIT Madras *Sept 2025*
Teaching Assistant, Electronic Systems Thinking & Circuits Lab — IIT Madras *May – June 2025*
Teaching Assistant, Embedded C Programming Lab — IIT Madras *May – June 2025*
Coordinator, Spark (ES Lab Activities) — IIT Madras *Jan 2025*
Group Leader, Kanha House — IIT Madras *Jan 2024 – Aug 2024*
Logistic Team Coordinator, Kanha House — IIT Madras *Oct 2023 – Jul 2024*

Teaching Assistant responsibilities included:

• Mentored 200+ students in embedded C programming, microcontroller debugging, and low-level memory optimization.

• Guided hands-on circuit design and system-level troubleshooting, improving lab performance and debugging efficiency.

EXTRACURRICULAR ACTIVITIES

• Member, Robosoccer Team — Paradox, IIT Madras (June 2024)
• Won Mathematics Aptitude Test at school level — Army Public School
• Member, IITM DIY School-Connect Program — IIT Madras (July 2024 – Jan 2025)
• 1st Place, State Level Science Drama Contest — State Council of Educational Research and Training (SCERT).